

long as no value reaches $p/2$ (the signed embedding, Vapor.Field). vapor's int8 contraction is already proved (Lean, integer parity) never to wrap 32-bit accumulators under its admissibility bound; the same bound, checked against $p/2$, makes the kernel's result and the circuit's the same number so the witness can be computed by the certified gemm i8 kernels on any substrate (witness/3 , certified: true) and the circuit merely checks it. Floating point, by contrast, is not field arithmetic: every rounding would cost a range proof. That is why f32 programs are not compiled here. Weights are constants of the circuit. A

product by a constant is free in R1CS (it is part of a linear combination): a hidden dense layer costs no constraint at all and the last one one per output; the cost of a proof is in the non-linearities (ReLU: a bit decomposition, $B + 3$ constraints per neuron) and in the input range check (9 per private int8 input). What is proved: there exists an int8 vector x with $\text{model}(x) = y$ for the public y the range check makes x int8 a fact of the circuit, and then every intermediate value is the integer the model computes (each stays below $p/2$ and below the ReLU decompositions' range, checked at compile time; Lean: field parity). Constants also